

Problem Statement

Problem Statement

A retail store manages different types of products. All products share common behaviours such as:

- Getting product name
- Getting product price
- Displaying product details

You are required to use an **interface** to model this system. Use the interface to implement three different classes:

- **Electronics**
Attributes – name, price, warranty (in months)
- **Grocery**
Attributes – name, price, expiryDate (String)
- **Clothing**
Attributes – name, price, size (String)

Input Format

First line: Category choice (integer)

- 1 → Electronics
- 2 → Grocery
- 3 → Clothing

Depending on the choice:

If choice = 1:

Name
Price
WarrantyMonths

If choice = 2:

Name
Price
ExpiryDate

If choice = 3:

Name
Price
Size

Constraints

- $0 < \text{price} \leq 10^7$
- $0 \leq \text{warranty} \leq 60$
- Names contain no spaces
- Expiry format: YYYY-MM-DD

Output Format

Print details in the **exact** format:

For Electronics:

Category: Electronics
Name: <name>
Price: <price formatted to 2 decimal places>
Warranty: <warranty> months

For Grocery:

Category: Grocery
Name: <name>
Price: <price formatted to 2 decimal places>
Expiry Date: <expiry>

For Clothing:

Category: Clothing
Name: <name>
Price: <price formatted to 2 decimal places>
Size: <size>

Sample Test Cases

Test Case	Sample Input	Sample Output
1	1 Laptop 50000 24	Category: Electronics Name: Laptop Price: 50000.00 Warranty: 24 months
2	2 Rice 2000 2026-12-31	Category: Grocery Name: Rice Price: 2000.00 Expiry Date: 2026-12-31
3	3 Shirt 1500 L	Category: Clothing Name: Shirt Price: 1500.00 Size: L

Hidden Test Cases (1 Mark each, Total = 5 Marks)

Test Case	Input	Expected Output
1	1 Phone 50000 12	Category: Electronics Name: Phone Price: 50000.00 Warranty: 12 months
2	2 Milk 60 2025-05-10	Category: Grocery Name: Milk Price: 60.00 Expiry Date: 2025-05-10
3	3 Jacket 3499 XL	Category: Clothing Name: Jacket Price: 3499.00 Size: XL

4	1 Mobile 9999.5 12	Category: Electronics Name: Mobile Price: 9999.50 Warranty: 12 months
5	3 Tshirt 500 M	Category: Clothing Name: Tshirt Price: 500.00 Size: M
